Ex.No.-2.2-Create a 3D solid model from isometric drawing.

AIM:

To create a 3Dmodel from the given isometric drawing using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

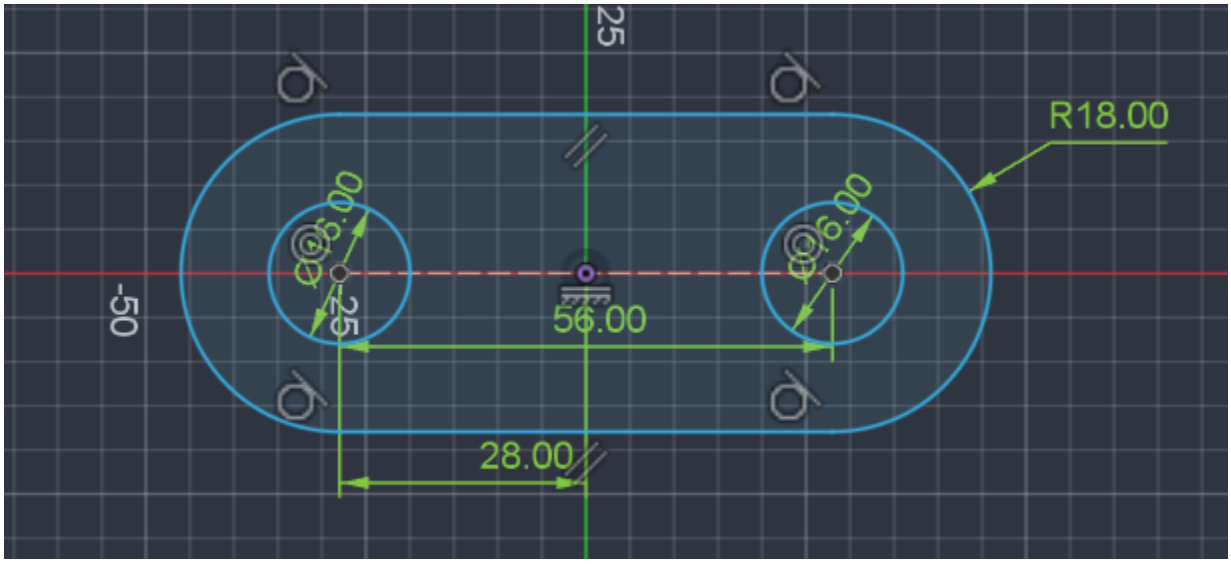
STEP-1:
Select the Sketch Plane
Select the TOP (XY) Plane

STEP-2:
Create a Sketch
Select Sketch > Rectangle > Center Rectangle.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the rectangle.
Type the Tab button and enter the values 56 mm and 18mm for the width and height respectively.

STEP-3:
Select Sketch > Arc > Center point arc.
Click again to place the value (Both end of the sketch). You should see the cursor snap to this location
Type the Tab button and enter the values R 18mm

STEP-4:
Select Sketch > circle > Center point circle.
Click again to place the value (Both side of the sketch). You should see the cursor snap to this location
Type the Tab button and enter the values dia 16mm

Click again to place the sketch.



Click Finish Sketch after completing.

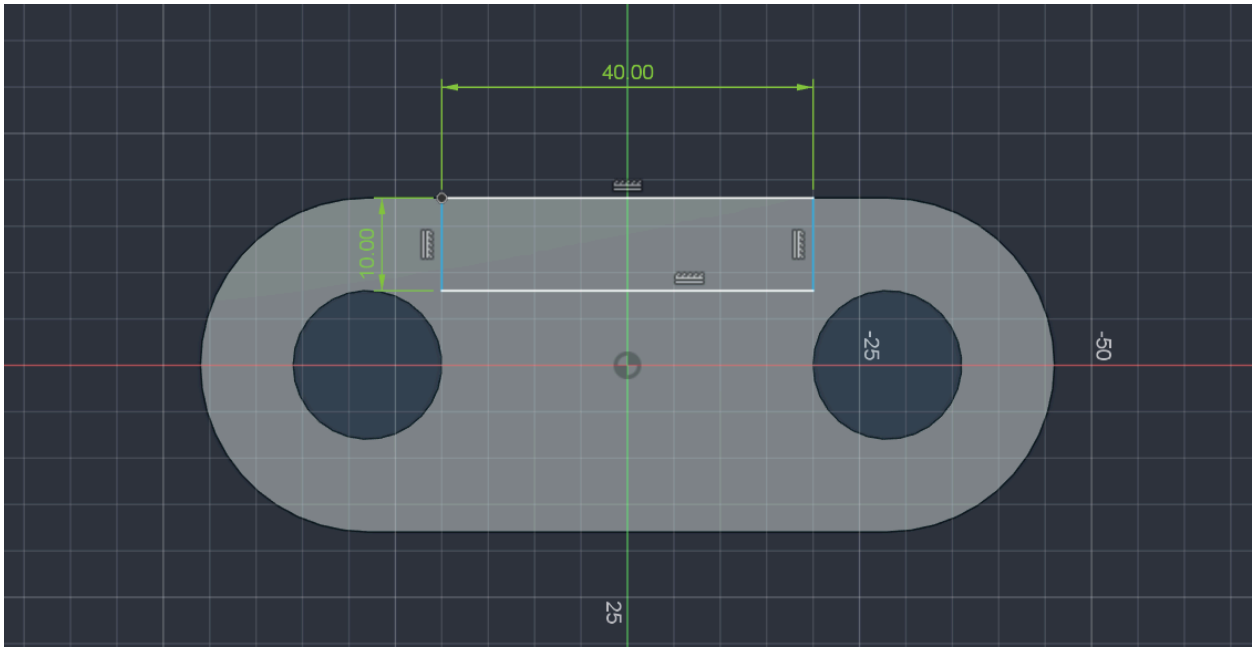
STEP-5:
Extrude Sketch in XY Plane
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of 8mm.
Press ENTER to accept the value.

STEP-6:
Start the sketch command
Select Sketch > Create Sketch.

STEP-7:
Select the Sketch Plane
Select the TOP SURFACE of the 3D block

STEP-8:
Create a RECTANGLE
Select Sketch > RECTANGLE > CENTER RECTANGLE.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the RECTANGLE.
Type the Tab button and enter the values 40mm and 10mm for the width and height respectively

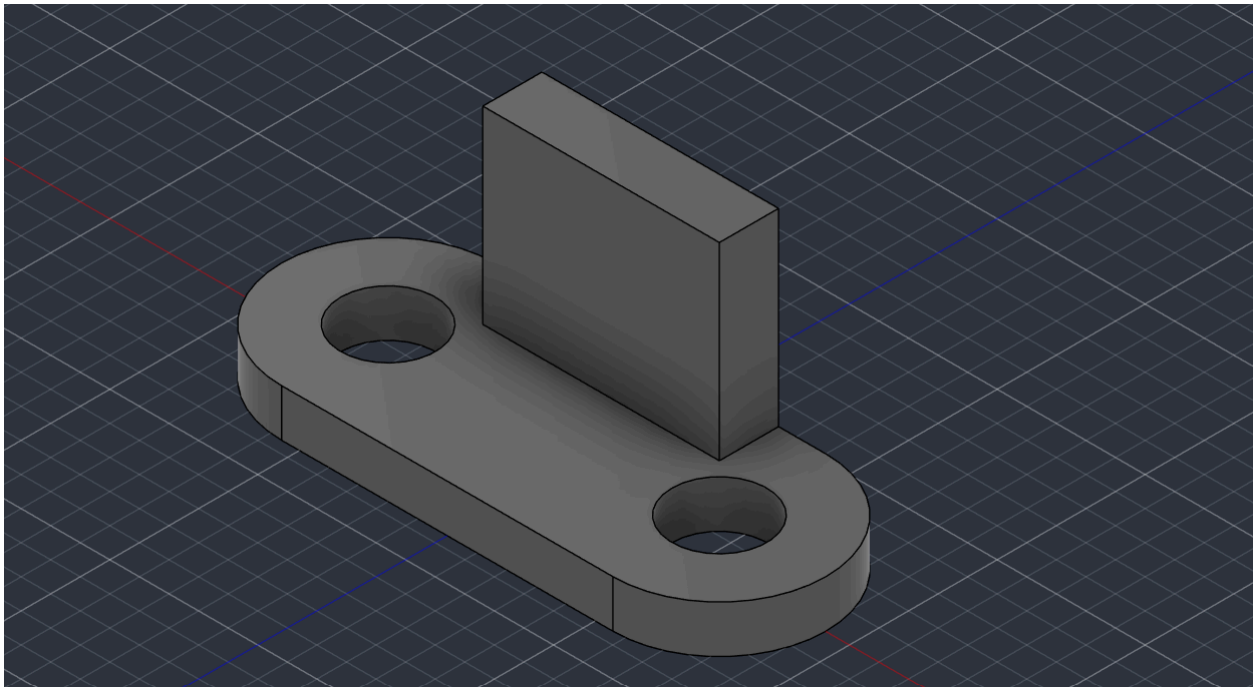
Click again to place the RECTANGLE.



Click Finish Sketch after completing.

STEP-9:
Extrude RECTANGLE
Select Extrude.
Select the Rectangle sketch.
Click again to place the value.
Type in the value of 32mm.

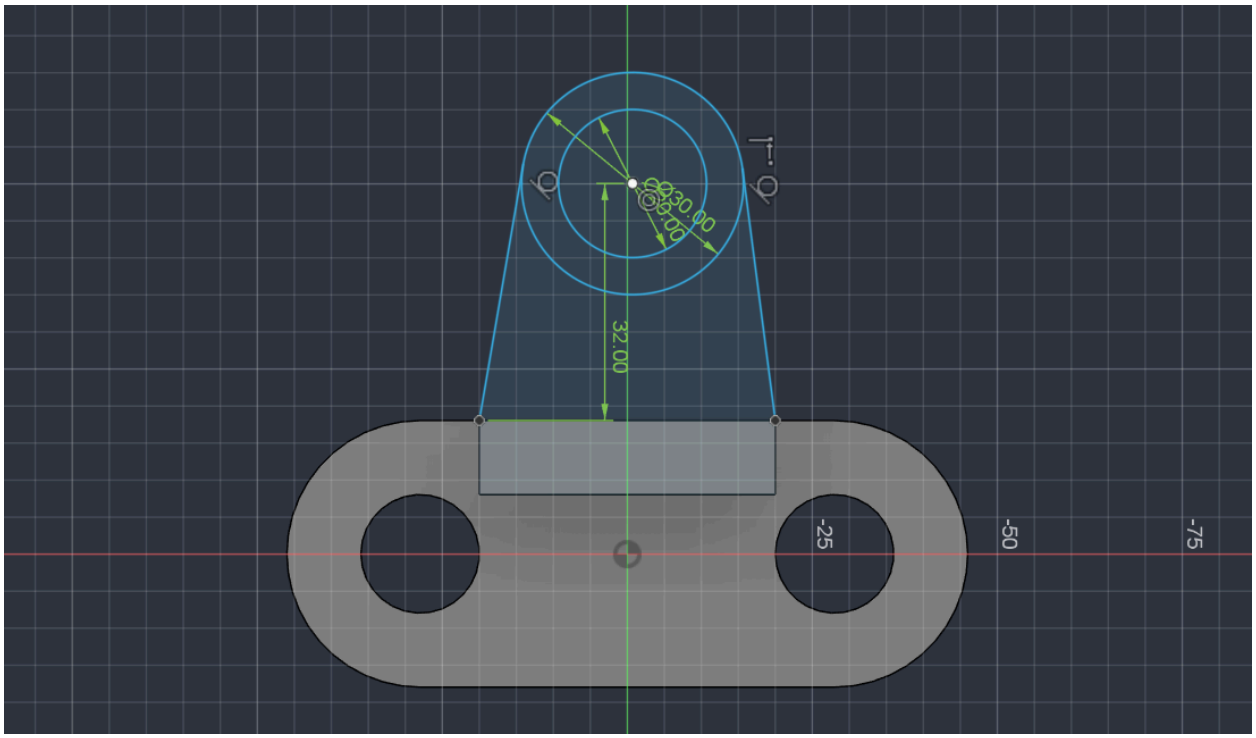
Press ENTER to accept the value.



STEP-10:
Select the Sketch Plane
Select the TOP SURFACE of the 3D block

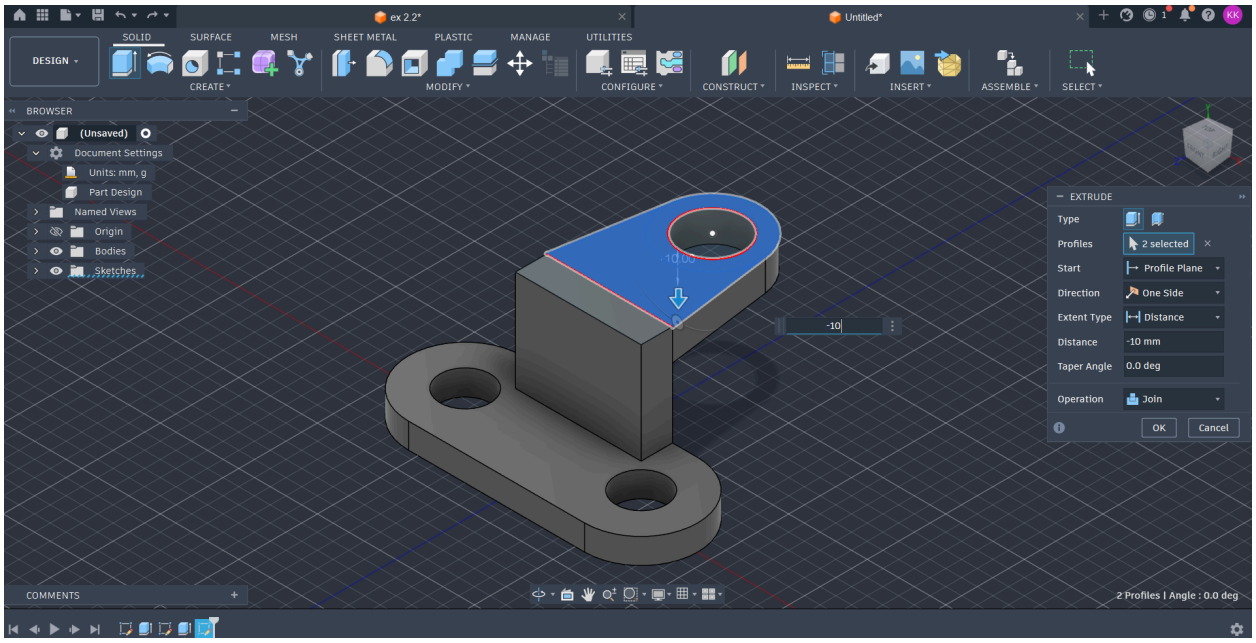
STEP-11:
Create two circles with same centre each of diameter 20mm and 30mm respectively.
They are placed at a distance of 32mm from the rectangle.

Two tangent lines are created such that they start at corners of the rectangle and connect to the circles.



Click Finish Sketch after completing.

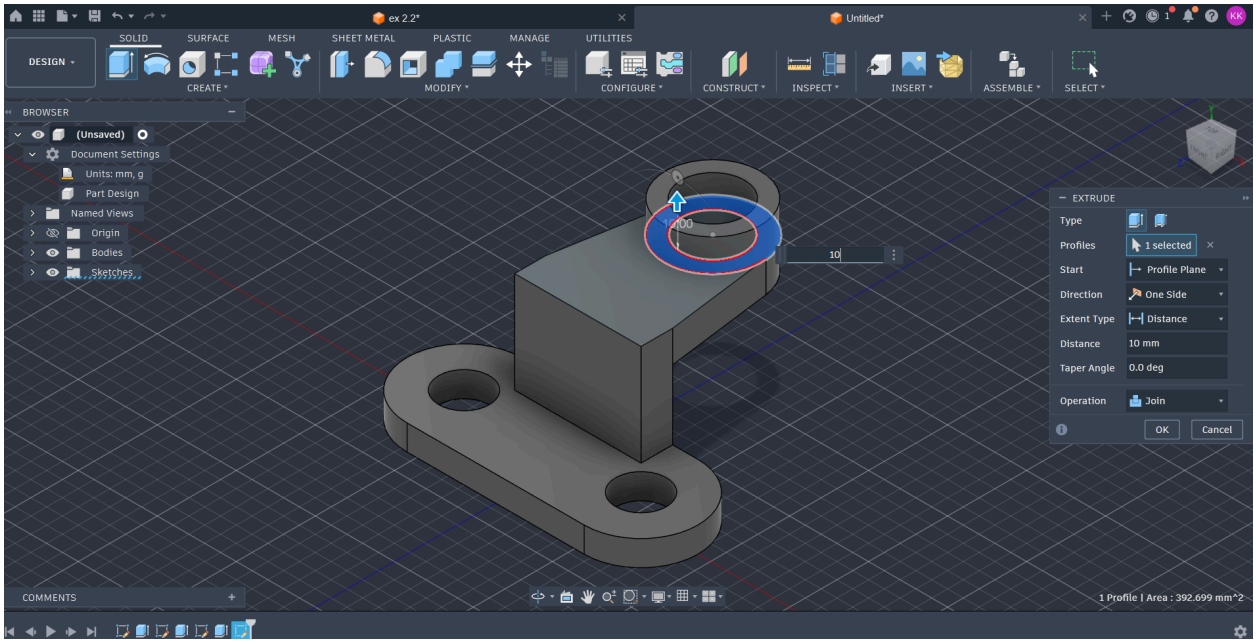
STEP-12:
Select Extrude.
Type in the value of -10mm.



Press ENTER to accept the value.

STEP-13:
Create a CIRCLE
Select Sketch > CIRCLE > CENTER CIRCLE.
Now hover over the origin of the sketch. You should see the cursor snap to this location.
Click once to begin placing the CIRCLE.
Type the Tab button and enter the value DIA 30mm and 20mm, inner and outer diameter respectively .
Click again to place the circle.
Click Finish Sketch after completing.

STEP-14:
Extrude CIRCLE
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of 10mm.

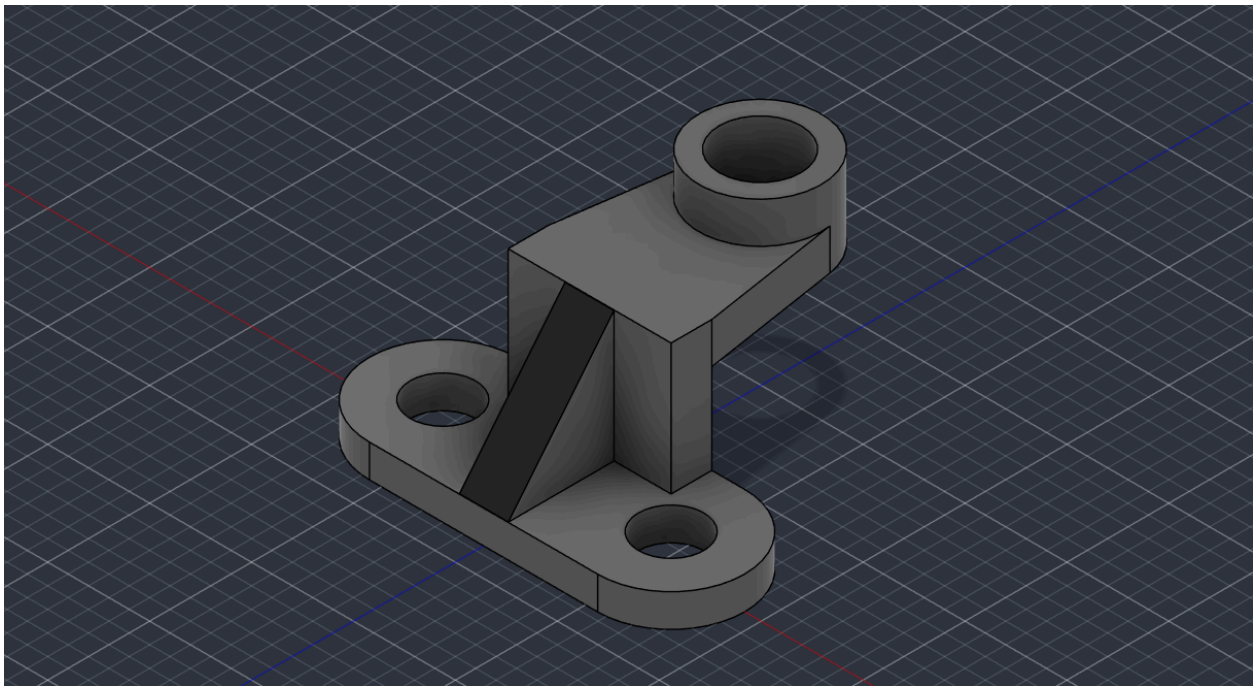


Press ENTER to accept the value.

STEP-15:
Start the sketch command
Select Sketch > Create Sketch

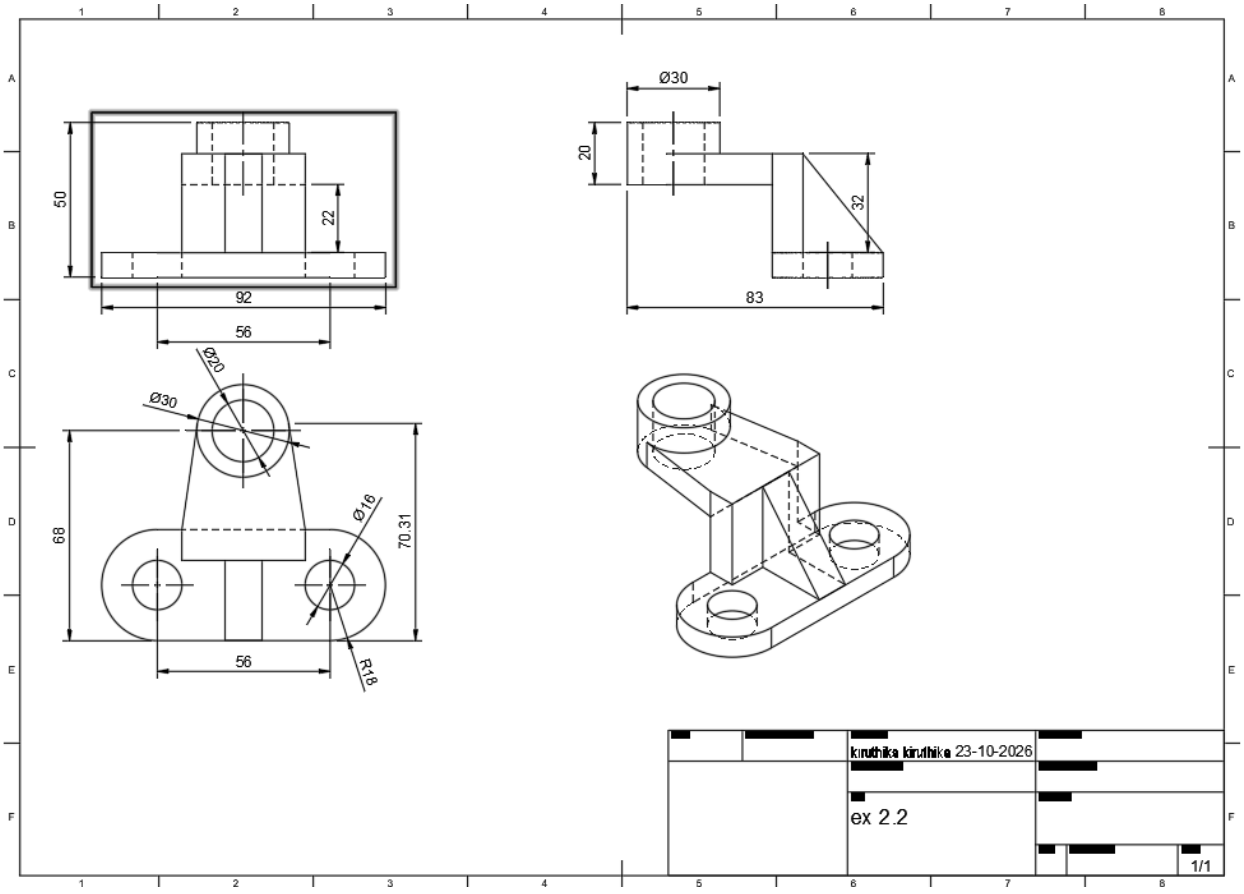
STEP-16:
Draw a line connecting the top and bottom.
Click Finish Sketch after completing.

STEP-17:
Select create > Create Rib.
Select the line.
Type in the value of 12mm.
Press ENTER to accept the value.



STEP-18:
Switch to drawing mode to plot the sketch in ISO standard.

OUTPUT:



RESULT:

Thus, the 3D model is created from given isometric drawing.